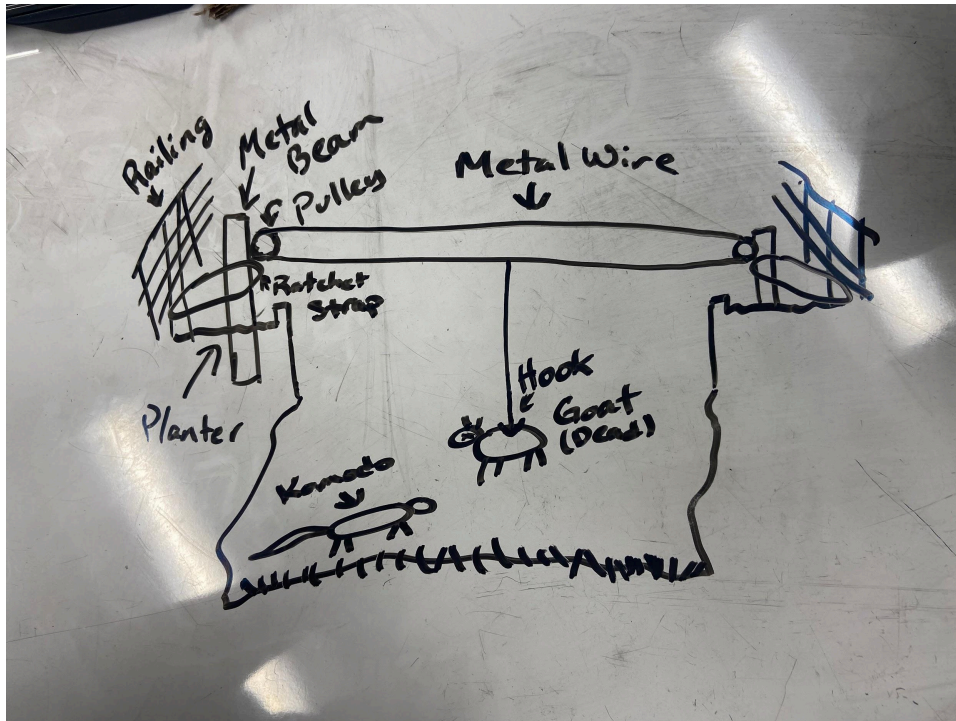


Element D

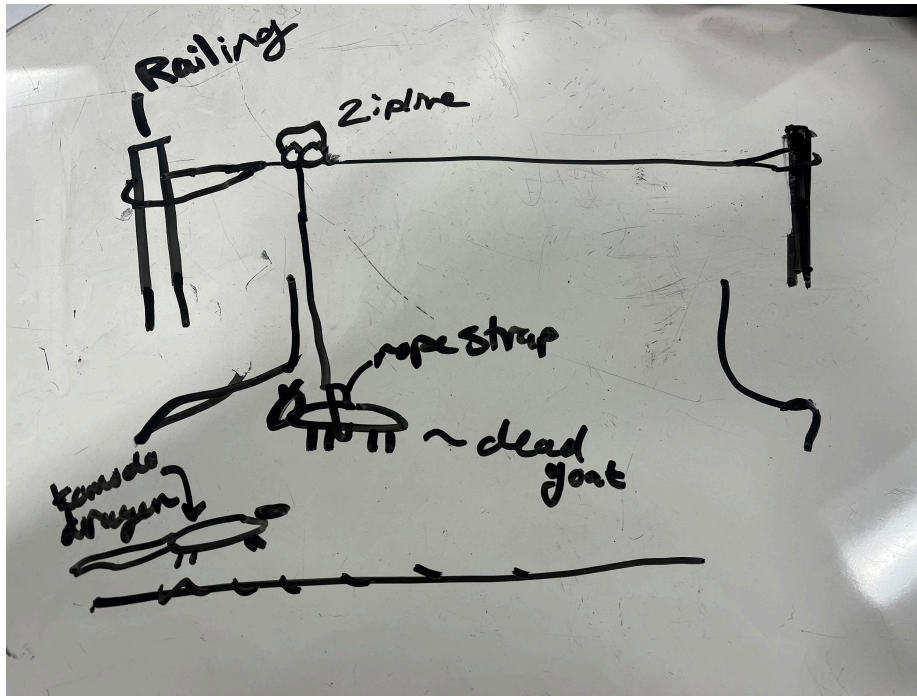
The zookeepers specifically asked for a pulley system to move the enrichment toy across the enclosure. However, we still have to design the actual toy that holds the meat.

Idea 1



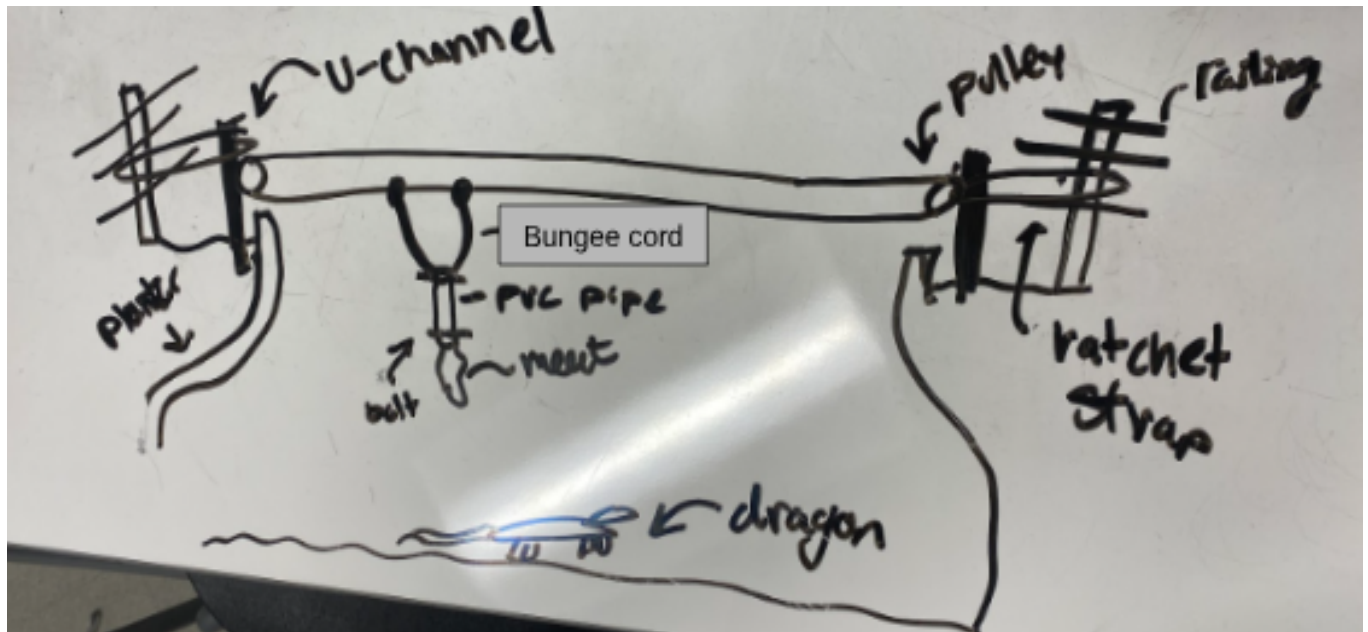
This system has a pulley spanning across the exhibit. The pulley is mounted in the planters with a metal U-channel. A ratchet strap pulls the U-channel to the railing close by, keeping it upright. The pulley is made from metal wire for strength. Attached to the pulley is a hook, which will hold the 50 lbs of meat at a height to where the komodo dragon can reach it.

Idea 2:



This system is a zipline that is meant to be pulled along 2 railings; the pulley is mounted in the planters with a metal U-channel. A ratchet strap pulls the U-channel to the railing close by, keeping it upright. There is the zipline track and attached to the length of the fire hose rope is a loop which can be adjusted to hold up the komodos dragon's food.

Idea 3



This system has a pulley spanning across the exhibit. The pulley is mounted in the planters with a metal U-channel. A ratchet strap pulls the U-channel to the railing close by, keeping it upright. The pulley is made from metal wire for strength. Attached to the pulley is 2 bungee cords that will hold a PVC pipe. This pipe has holes along the bottom, and bolts that screw into it. The meat will get screwed in, and it can be pulled after a couple different runs.

Criteria	Weight	Solution A (Hook and Ratchet Strap)		Solution B (net)		Solution C (Pipe)	
		Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Height	20%	4	0.8	3	0.6	4	0.8
Food Detachabil	20%	2	0.4	4	0.8	4	0.8
Length	20%	5	1	5	1	5	1
Speed	18%	3	0.54	5	0.9	3	0.54
Ability to hold 50	18%	3	0.54	2	0.36	4	0.72
Water resistant	4%	3.5	0.14	4.5	0.18	4	0.16
SUM	100%		3.42		3.84		4.02

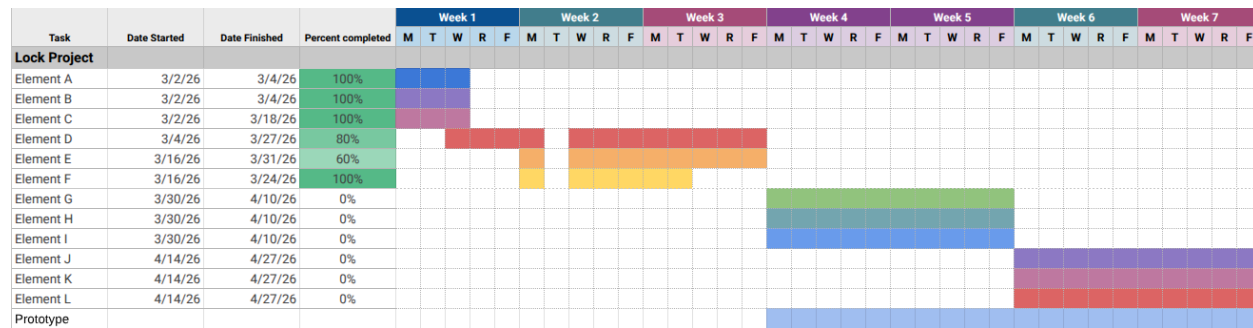
Based on the decision matrix, we are choosing Solution C. This will allow us to keep the attachment out of the dragon's reach while also holding the carcass in place.

List Chart w/ Prices of Items and Number Needed

Item	Cost	# needed	total
Aircraft cable	\$76.06	1	\$76.06
Pulley (2 pack)	\$17.77	1	\$17.77
Ratchet strap	\$22.00	2	\$44.00
Unistrut U-channel	\$21.70	2	\$43.40
Bungee cord	\$25.00	1	\$25.00

The sum of the supplies needed: \$206.23

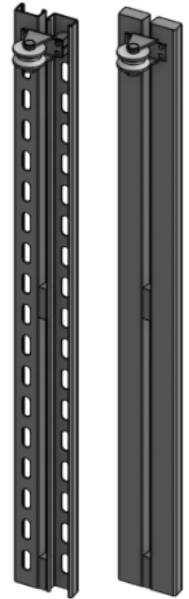
Gantt Chart:



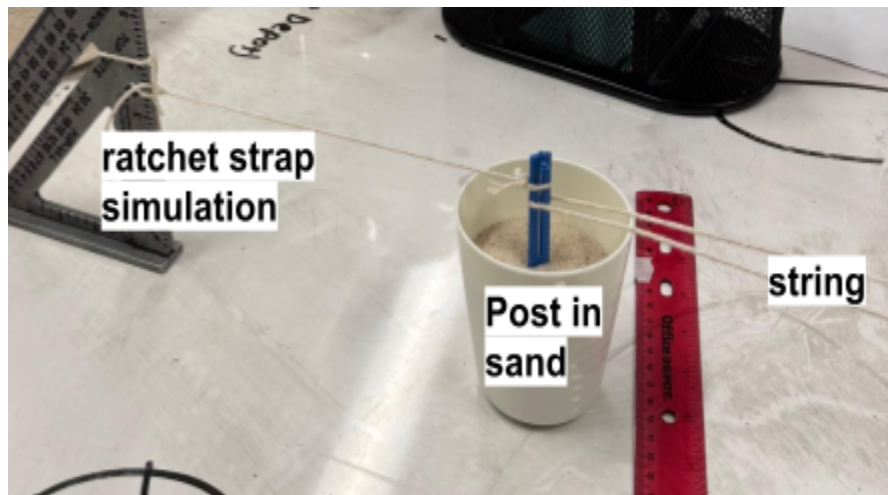
Model:

Since our project is very large, we decided to create a scaled-down model to mathematically check our design on whether it will hold the correct amount of weight.

In CAD, we designed our posts, with a pulley and the two U-channels, to scale. The pulley will be screwed into the holes in the U-channel. However, we had to change this design for 3D printing. Since a scaled-down version of this design creates sizes too small for the printer, we had to change the U-channels to be a solid block, as demonstrated in the picture on the right. This made the process of creating the model easier.

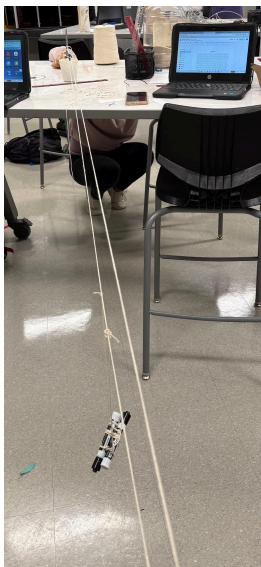
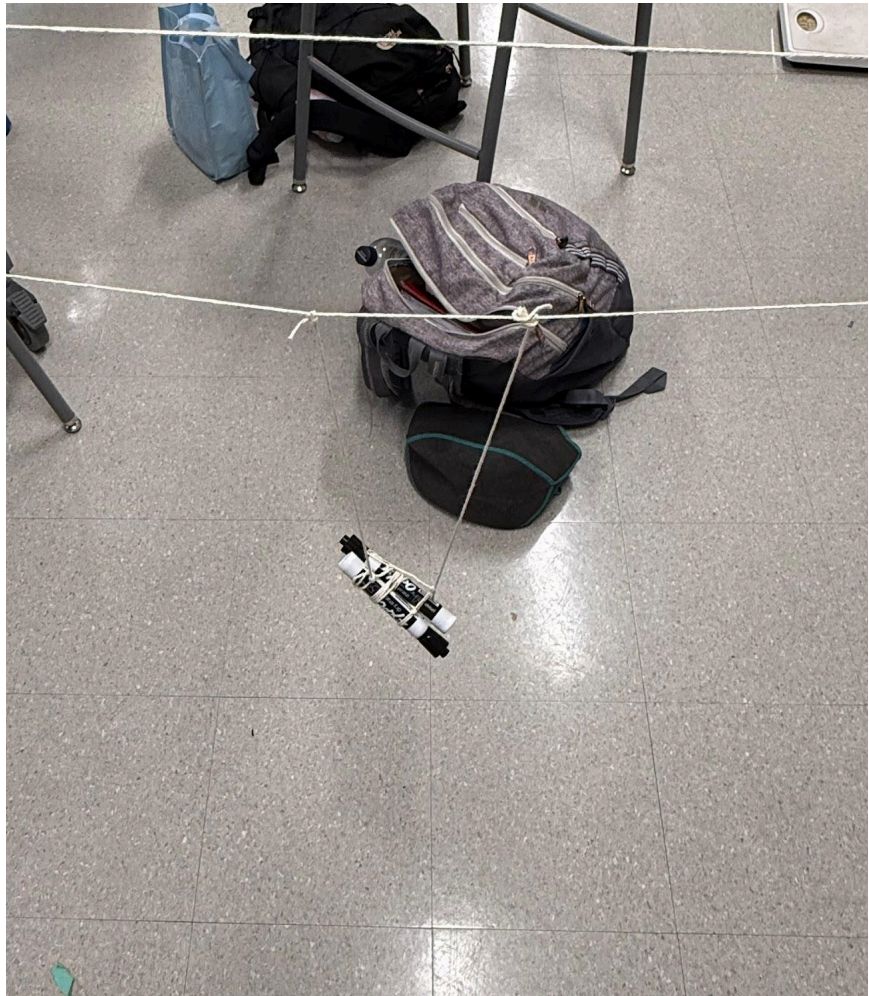


After this, we printed the model at a 1/10 scale. This allows for easy calculations and a manageable model to do in our section of the classroom. The calculated distance between the posts on the model is 105.6 inches, which is 1/10 of the total 88ft distance.



When making the model, we used a small cup with wet sand to represent the planters that the posts will be placed in. The triangle with string is a representation of the ratchet strap from the post to the fence. The string was mainly tied using a two-half hitches knot.

After making the posts, we tied the string tightly between the posts. After this, we made an apparatus to simulate the bungee cords and the overall weight of the attachment-carcass system. From our calculations, we calculated that the model should be able to hold 0.05lbs or 22.68 grams. The two expo markers weigh approximately 25.5 grams, thus being heavier than the necessary weight. We attached the markers together using clove hitch knots, and then using more string to attach it to the center of the string. This is the worst possible scenario in that the markers exert the most amount of torque on the posts.



Overall, through our creation of the model, we can conclude that the model will hold the necessary amount of weight, but there will be bowing due to the weight of the wire and the attachment.